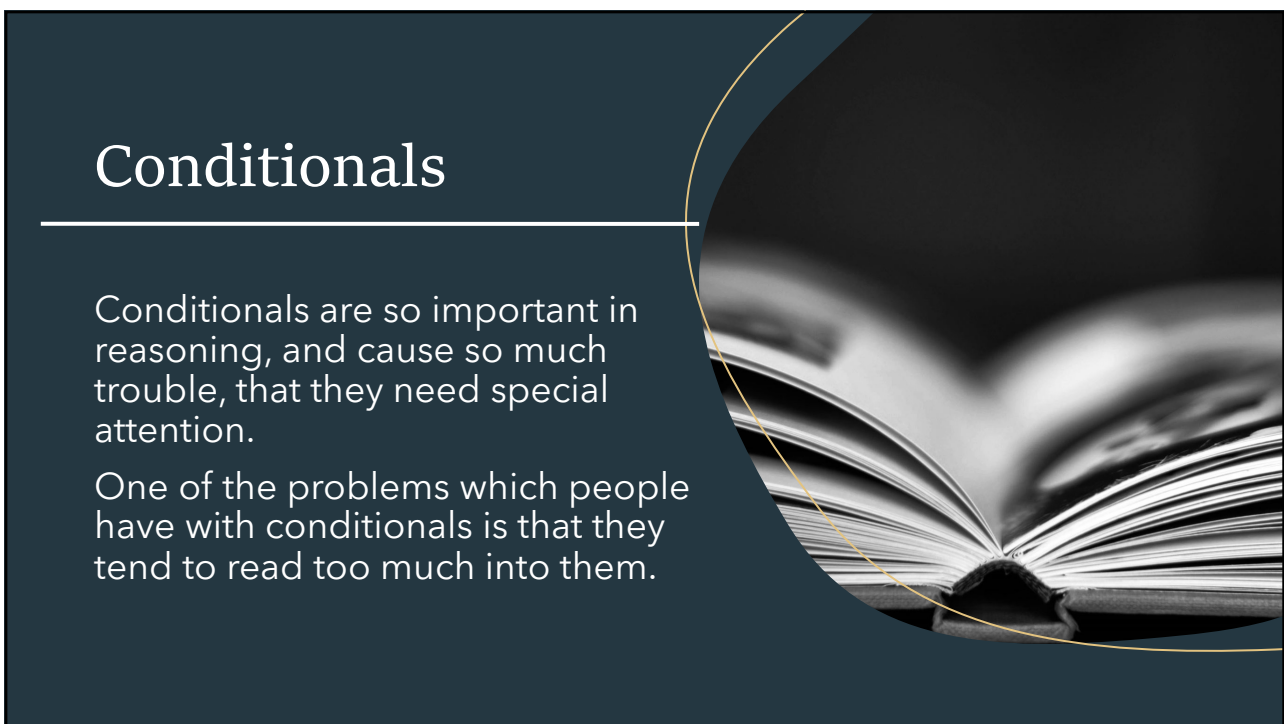




1



2

Consider this example:

You have four cards, each with a letter of the alphabet on one side and a number on the other.

You can see only one side of each card, and they look like this:

E K 6 7

Here is an hypothesis about what is written on the cards:

If a card has a vowel on one side, then it has an even number on the other.

Which card (or cards) if turned over might show that the hypothesis about what is written on the cards is false?

3

Another example...

If Harry Lee is a member of the Che Nghee Khor Moral Uplifting Society, then he is Chinese

Form a new conditional from that one by exchanging antecedent and consequent and then negating both:

If Harry Lee is not Chinese, then he is not a member of the Che Nghee Khor Moral Uplifting Society

The move is known as *transposition* and is deductively valid.



4

The fallacy of conversion...

What about the conditional which is formed from the original by exchanging antecedent and consequent **without** adding the negations?

This is called the original conditional's *converse*.

They are **not** logically equivalent.

If Harry Lee is Chinese, then he is a member of the Che Nghee Khor Moral Uplifting Society

That is a ***fallacy of conversion***.



5

What's the answer?

You have four cards, each with a letter of the alphabet on one side and a number on the other.

E K 6 7

Here is an hypothesis about what is written on the cards:

If a card has a vowel on one side, then it has an even number on the other.

Which card (or cards) if turned over might show that the hypothesis about what is written on the cards is false?



6

Transposition...

a **deductively valid** move taking us from this conditional:

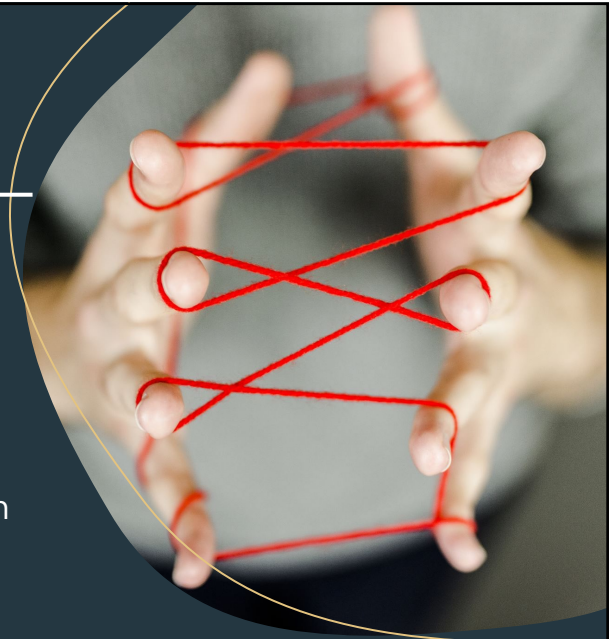
If A then B

to this one:

If not-B then not-A

You can do a further transposition on this new conditional:

If not-not-A then not-not-B



7

Double negation

Consider this example:

Cedric is strange

Cedric is not strange

It is not the case that Cedric is not strange

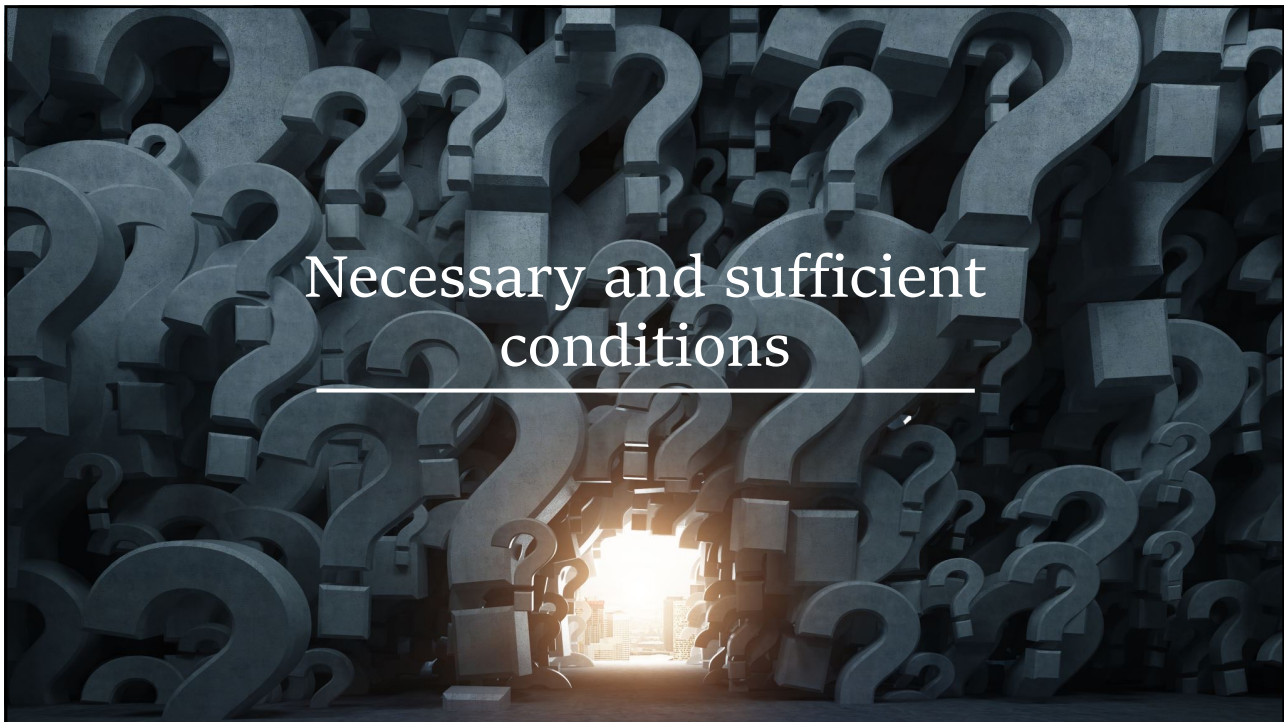
What can this last sentence mean except that Cedric is strange?

We can generalise this observation to form the principle:

Double Negation: A is logically equivalent to not-not-A



8



9

Necessary and sufficient conditions

In a conditional statement, the antecedent is claimed to be sufficient for the consequent.

We can generalise:

'If (antecedent), then (consequent)'

The antecedent is sufficient for the consequent; and the consequent is necessary for the antecedent.



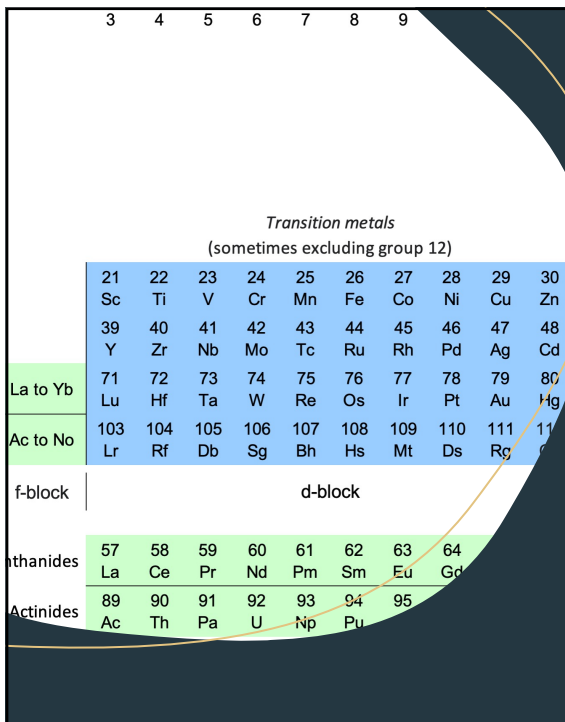
10

An example

If iron is heavier than tin, then Mendeleev's Periodic Table of the elements is incorrect.

Iron's being heavier than tin would be **sufficient** for the incorrectness of the Periodic Table

Less obviously, the incorrectness of the Periodic Table would be a **necessary** condition for iron's being heavier than tin.



		3	4	5	6	7	8	9	
		Transition metals (sometimes excluding group 12)							
		21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni
		39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd
La to Yb		71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt
Ac to No		103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds
f-block		d-block							
Lanthanides		57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd
Actinides		89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	

11

An example...

"Iron is heavier than tin" is sufficient for "The Periodic Table is incorrect"

and

"The Periodic Table is incorrect" is necessary for "Iron is heavier than tin".

Iron couldn't be heavier than tin without the Periodic Table's being incorrect.



12

'Only if'

Harry Lee is eligible to be a member of the Che Nghee Khor Moral Uplifting Society only if he is Chinese

This means the same as this:

If Harry Lee is eligible to be a member of the Che Nghee Khor Moral Uplifting Society, then he is Chinese

Being Chinese is a **necessary condition** of belonging to this society.

The 'only if' version of the conditional is the more natural way to express a necessary condition.



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'Only if'

'Only if' statements are tricky because the bit which follows the 'if' in them is in fact the **consequent** of the normal conditional.

In the normal conditional itself, the bit that follows the 'if' is the antecedent.

In general:

'If (antecedent), then (consequent)' = '(Antecedent) only if (consequent)'

14



'Unless'

This conditional:

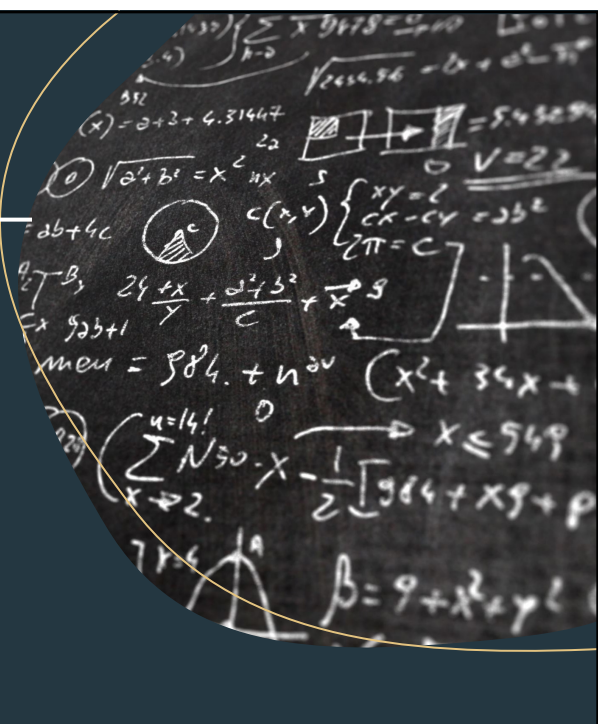
If Harry Lee is eligible to be a member of the Che Nghee Khor Moral Uplifting Society, then he is Chinese

Means the same thing as this:

Harry Lee isn't eligible to be a member of the Che Nghee Khor Moral Uplifting Society unless he is Chinese

'If P then Q' is equivalent to ***'Not-P unless Q'*** and to ***'P only if Q'*** and to ***'If not-Q then not-P'***

15



In general

'If P then Q'
is equivalent to:
'Not-P unless Q'
and
'P only if Q'
and
'If not-Q then not-P'

16

'If and only if'

Where you want to express a pair of conditions which are necessary and sufficient for each other, you must employ both the 'if' **and** the 'only if' e.g.:

An act is a murder if and only if it was unlawful killing with malice aforethought.

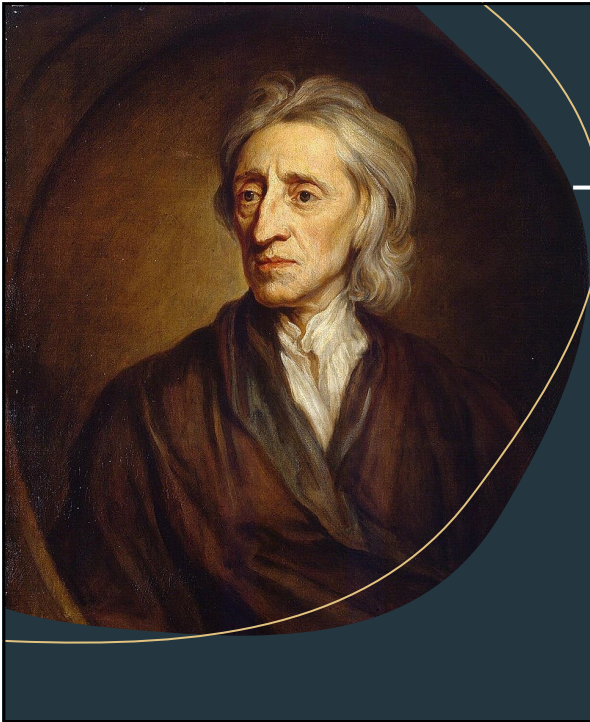
The expression of necessary **and** sufficient conditions can often correspond to the definition of a term.

Most common shortenings are 'iff', 'just if', and 'just in case'.

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Conditionals are tricky...

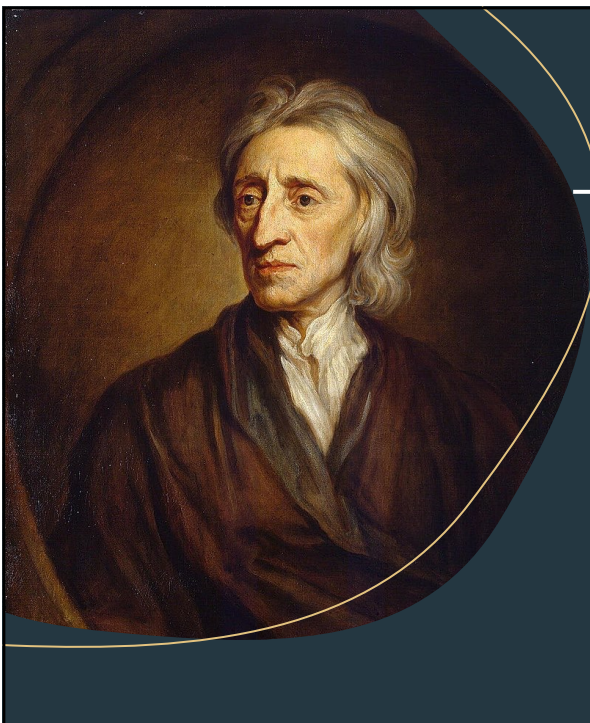
18



Locke's mistake...

The materials of nature (air, earth, water etc.) that remain untouched by human effort belong to no one and are not property. It follows that a thing can become someone's private property only if he works and labours on it to change its natural state. From this I conclude that whatever a man improves by the labour of his hand and brain belongs to him, and to him only.

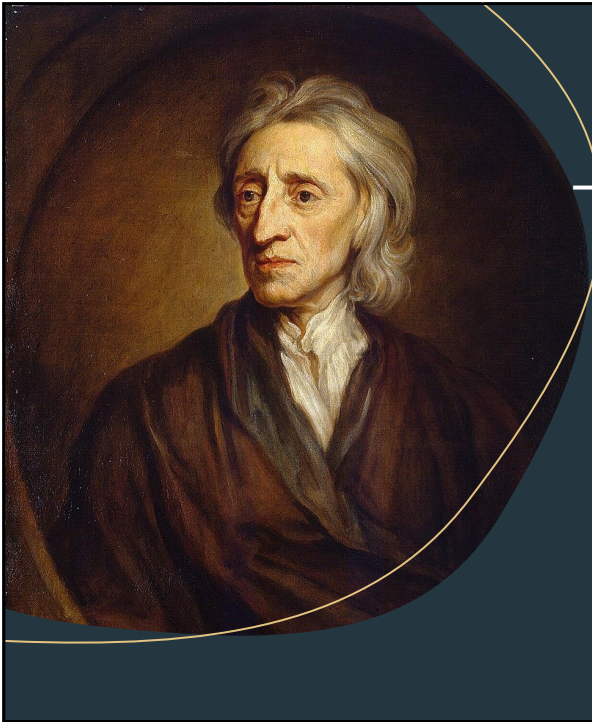
19



Locke's mistake...

The materials of nature (air, earth, water etc.) that remain untouched by human effort belong to no one and are not property. It follows that a thing can become someone's private property only if he works and labours on it to change its natural state. From this I conclude that whatever a man improves by the labour of his hand and brain belongs to him, and to him only.

20



Locke's mistake...

(1)<The materials of nature (air, earth, water etc.) that remain untouched by human effort belong to no one and are not property>. It follows that (2) <a thing can become someone's private property only if he works and labours on it to change its natural state>. From this I conclude that (3) <whatever a man improves by the labour of his hand and brain belongs to him, and to him only>.

21

(1)<The materials of nature (air, earth, water etc.) that remain untouched by human effort belong to no one and are not property>. It follows that (2) <a thing can become someone's private property only if he works and labours on it to change its natural state>. From this I conclude that (3) <whatever a man improves by the labour of his hand and brain belongs to him, and to him only>.

The first statement is a disguised conditional, which says roughly:

(1) If some natural material is not worked on, then it is not property

or, more abbreviated still:

(1) If not worked on, then not property

22

(1) <The materials of nature (air, earth, water etc.) that remain untouched by human effort belong to no one and are not property>. It follows that (2) <a thing can become someone's private property only if he works and labours on it to change its natural state>. From this I conclude that (3) <whatever a man improves by the labour of his hand and brain belongs to him, and to him only>.

Locke's move from (1) to (2) is deductively valid.

First, transpose the original conditional

(1) If not worked on, then not property

to get this:

(1*) If not not property, then not not worked on

Then remove the two negations via double negation:

(1) If property, then worked on**

23

(1) <The materials of nature (air, earth, water etc.) that remain untouched by human effort belong to no one and are not property>. It follows that (2) **<a thing can become someone's private property only if he works and labours on it to change its natural state>**. From this I conclude that (3) <whatever a man improves by the labour of his hand and brain belongs to him, and to him only>.

(1) If property, then worked on**

Transform this new conditional into its 'only if' version:

(2) Property only if worked on

(2) states a **necessary** condition of something's being property.

But Locke's conclusion (3) is another disguised conditional.

(3) If worked on, then property

(3) states a **sufficient** condition for something's being property.

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Locke's mistake...

Locke has apparently got confused.

There may be other necessary conditions than the one he mentions for some natural material's being property.

The moral(s) of the story:

You don't get a sufficient condition from a list of necessary conditions unless you also know the list of necessary conditions is complete.

Conditionals are tricky.



25

Argument forms and conditionals

There are some standard argument forms employing conditionals, some of which are deductively valid and some of which are not.

Logic books tend to hold up the deductively valid forms as models of correct argumentation and to stigmatise the others as fallacious.

Matters are more complicated than this.



26

Hypothetical syllogism

If A then B

If B then C

Therefore

If A then C

The validity of this form should be obvious.

It makes no difference to the validity of the inference what the letters stand for.

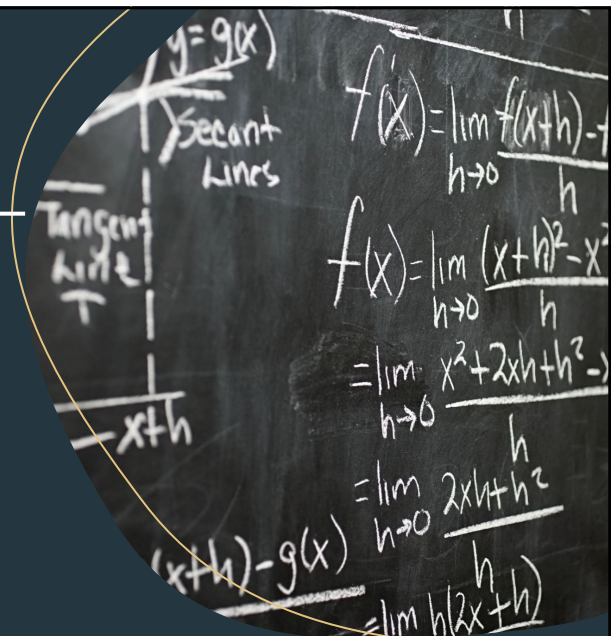
27

Modus Ponens & Modus Tollens

Modus Ponens: If A, then C,
A
Therefore, C.

Modus Tollens: If A, then C
Not-C
Therefore, Not-A.

Any argument that takes either of these forms will be deductively valid.



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But remember...

...the fact that you have a deductively valid argument is no guarantee that you have an argument which proves its conclusion.

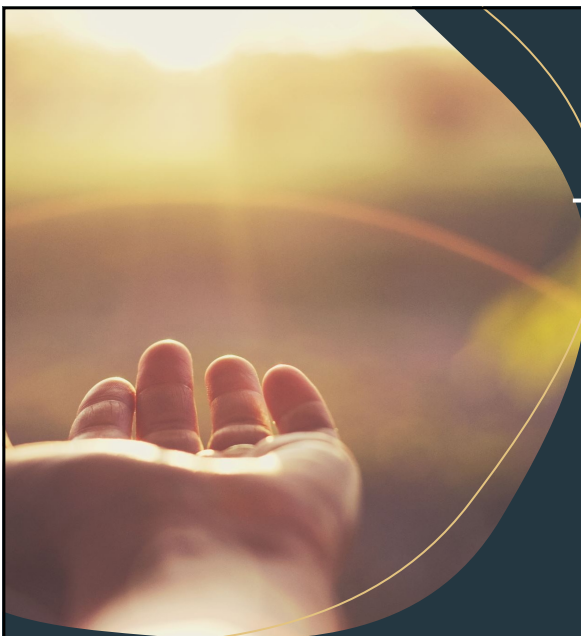
If Gary is a liar, then he does not always tell the truth

Gary is a liar

So

Gary does not always tell the truth

29



Question-begging?

If God loves everyone, then God loves you

God loves everyone

So

God loves you

In some contexts, this argument would not beg the question.

30

Two invalid forms...

Denying the antecedent: If A, then C

Not-A

Therefore, Not-C

Affirming the consequent: If A, then C

C

Therefore, A

There is no doubt that arguments of these forms are not deductively valid.

31

Invalid forms are tempting...

*If Sarah discriminates against men, she will have
given bad marks to all her male students this year.*

*But Sarah has given bad marks to all her male
students this year.*

So

She discriminates against men.

32

Invalid but not necessarily fallacious...

If Sarah discriminates against men, she will always give bad marks to her male students.

But in fact she always does give bad marks to her male students.

So

She discriminates against men.

Both of these arguments affirm the consequent, and neither is deductively valid BUT I wouldn't want to call the second one fallacious.

33

The moral of the story...

A knowledge of which forms of argument are valid and which are invalid is essential both for evaluation and construction of arguments

BUT

An argument with an invalid form may still be a very strong non-deductive argument.

And even if you construct an argument that has a valid form you can still commit other sorts of fallacies like begging the question.

34

Conditionalisation

For every argument, there corresponds a conditional which has as its antecedent the conjunction of the premises and as its consequent the conclusion.

Suppose the premise is P, and the conclusion C, and the argument from P to C:

P Therefore, C

The corresponding conditional in this case would be:

If P then C

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Conditionalisation

The conditional and the argument are **NOT equivalent** but they **correspond**.

The letter 'P' can stand for as many premises as you like, since a set of premises is equivalent to a conjunction which contains all those premises as conjuncts.

If the argument had been:

[P1 and P2 and P3 and P4], so C

then the corresponding conditional would have been

If (P1 and P2 and P3 and P4), then C

36

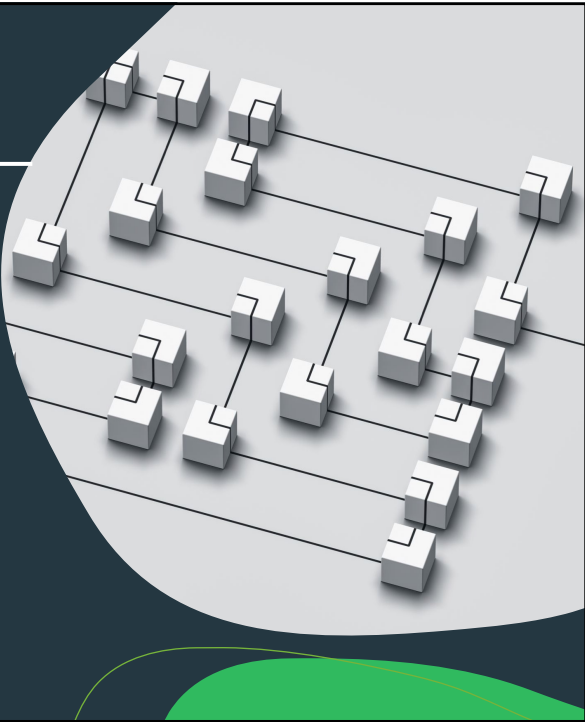
Conditionalisation

When we form the conditional corresponding to an argument, we abandon the assertion of the premises and conclusion.

Instead, we assert that there is a connection between those premises and that conclusion.

That means that if there is such a connection, then the corresponding conditional will be true.

If there isn't that connection, then the corresponding conditional will be false.



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Conditionalisation

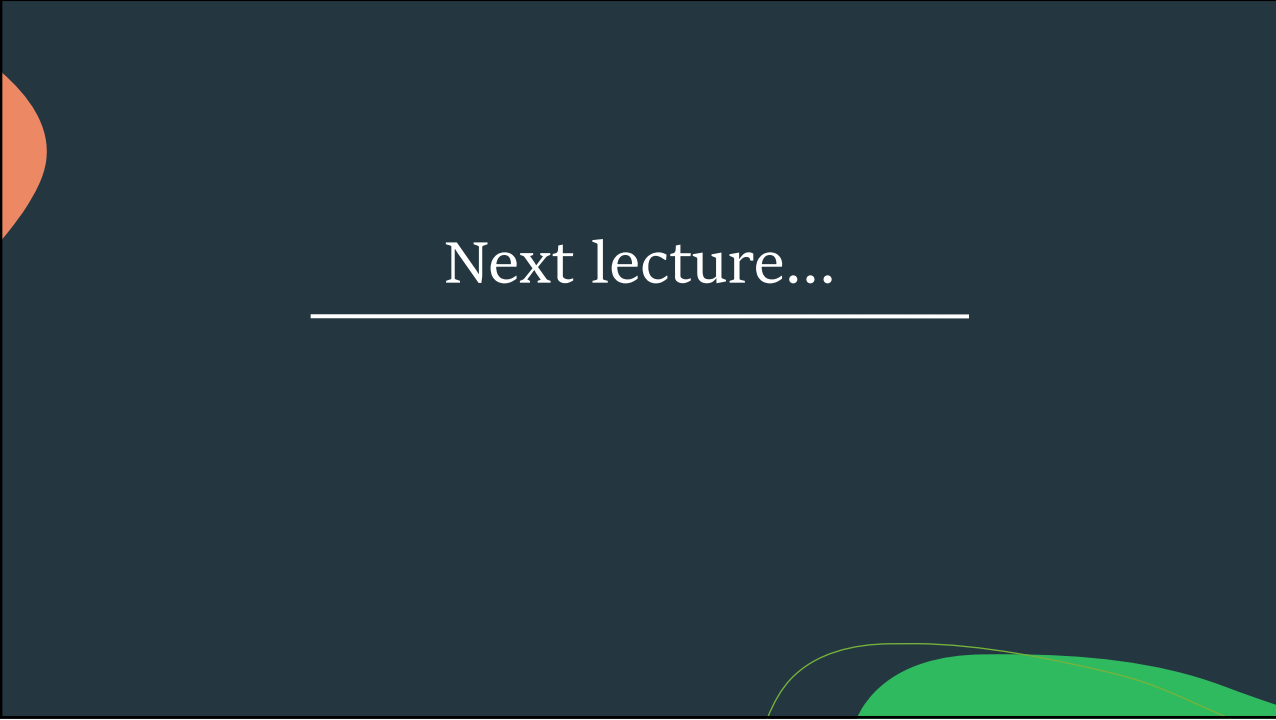
When the degree of support between premises and conclusion is at its maximum (valid) then the corresponding conditional is not only true, it is **necessarily** true e.g.:

This ball is red, Therefore, This ball is coloured

Corresponding necessarily true conditional:

If this ball is red, then it is coloured

38



Next lecture...